Get Fit with MA?
Estimating the Impact of Medicare Advantage on Physical Activity

Jong Min Jung* Jing Li†

Abstract

Medicare Advantage (MA) plans increasingly offer fitness benefits, yet the evidence on their behavioral effects remains limited. This paper provides quasi-experimental evidence on whether MA enrollment increases physical activity among older adults Exploiting the Medicare eligibility threshold at age 65, we employ a difference-in-discontinuities design comparing the discontinuous change in daily vigorous exercise between MA enrollees and non-MA individuals using nationally representative panel data from Health and Retirement Study. We find that MA enrollment increases the probability of daily vigorous exercise by approximately 4 percentage points. The estimates remain stable across alternative bandwidths and we do not find any pre-trends. The behavioral response is concentrated among groups facing the greater structural barriers to exercise. Non-White beneficiaries increases their daily workout frequency approximately by 14 percentage points, while women and non-college educated seniors show meaningfully larger effects than Non-MA counterparts. The effect attenuates within two years, suggesting one-time benefit access is insufficient for sustained habit formation. These findings demonstrate that insurance benefit design can shape preventive health behavior, highlighting supplemental fitness benefits as an important policy margin for promoting healthy aging

Keywords: Medicare Advantage, physical activity, fitness benefits, Difference-in-Discontinuities

*Department of Economics, University of Washington
†Department of Pharmacy, University of Washington

1 Introduction

As the U.S. population ages, promoting healthy behaviors among seniors has become both a public health imperative and an economic necessity. Medicare accounts for 21% national health expenditure and 10% of federal budget (Cubanski and Neuman (2023)), and these financial burdens will only intensify as adults over 65 outnumber children under 18 for the first time by 2034 (U.S. Census Bureau (2018)). At the same time, even modest improvements in health behaviors at older ages can yield substantial returns in longevity and quality of life, ultimately lowering health care utilization (Fries et al. (2011)). Among preventive behaviors, physical activity stands out as it is one of the most cost-effective interventions to promote healthy aging but it remains severely underutilized.

Physical inactivity is pervasive among older adults in the United States. More than 25 percent of adults aged 65 and older report no leisure-time physical activity, and only 14 percent meet both the aerobic and muscle-strengthening components of the 2018 Federal Physical Activity Guidelines (CDC (2022)).These guidelines recommend at least 150 - 300 minutes of moderate-intensity or 75-150 minutes of vigorous-intensity exercise per week, along with muscle-strengthening activities on two or more days. Especially for older adults, the guidelines recommend the exercise should include multicomponent physical activity, particularly the balance training. Compliance varies substantially across demographic groups and regions. Only 12 percent of older adults in the South meet the combined standard compared with 17 percent in the West, and seniors in metropolitan area are considerably more active than those in non-metropolitan areas. areas(Elgaddal and Kramarow (2024)).

The consequences of physical inactivity among seniors are severe. Insufficient physical activity increases the risk of cardiovascular disease, diabetes, cognitive decline, musculoskeletal disorders (Warburton et al. (2006);Lee et al. (2012)), and is a leading contributor to falls which is the primary cause of injury-related death among seniors (Jiang and Wang (2022)).

The economic burden is equally substantial as Medicare expenditures attributable to insufficient physical activity have been estimated at $117 billion annually in the U.S.(Carlson et al. (2015)). Physical inactivity is therefore not merely a clinical concern but a first-order fiscal challenge for the nation.

Physical inactivity also reflects and reinforces broader gaps in helath outcomes across demographic groups. Structural barriers such as financial constraints, limited access to recreational facilities, lack of transportation, and insufficient social support disproportionately affect low-income, ethnic minorities, and less-educated populations(Elgaddal and Kramarow (2024)). These disparities imply that the populations most likely to benefit from increased physical activity are those least likely to engage in it, raising an important policy question about whether interventions embedded in insurance benefit design can reduce structural barriers to exercise and narrow the gaps in physical activity among older adults.

Medicare Advantage(MA), the privately administered alternative to Traditional Medicare(TM), offers one promising lever. Unlike TM, most MA plans provide supplemental free fitness benefits such as SilverSneakers, Renew Active, Silver&Fit. These fitness programs provide free or subsidized gym memberships, instructor-led group exercise classes, and community-based fitness opportunities, which directly target the financial and logistical barriers to exercise among older adults. The availability of these benefits has grown rapidly. By 2025, approximately 95 percent of individual MA plans offer a free fitness benefit, and MA enrollment now covers more than half of all Medicare beneficiaries, with that share projected to increase to 64 percent by 2033 (KFF (2025)). Nevertheless, rigorous causal evidence on whether MA fitness benefits change exercise behavior remains limited.

Existing evidence on the behavioral impact of MA fitness programs is largely descriptive. Studies of program participants report improvements in physical functioning, emotional and mental well-being, mobility, , lower inpatient admissions and self-reported healthcare expenditures(Belza et al. (2006); Hamar et al. (2013) ; Brady et al. (2018) ; Nguyen et al. (2008)). However, these studies examine individuals who are already participating in the programs, raising concerns about selection on motivation, baseline health, and other determinants of exercise behavior(Newhouse (2014)). More broadly, a related literature uses the age-65 Medicare eligibility threshold to study the effects of Medicare on healthcare utilization, mortality and financial risks. Card et al. (2008) and Card et al. (2009) show that Medicare eligibility increases healthcare use and imporves outcomes for seriously ill patients, while Barcellos and Jacobson (2015) document that Medicare reduces exposure to meidcal expenditure risk. On the Medicare plan choice margin, MA enrollees have shown to differ systematically from TM enrollees, highlighting the presence of selection in plan choice(Abaluck and Gruber (2011); Abaluck and Gruber (2016) ; Cabral et al. (2018)). Taken together, these studies establish both the policy relevance of Medicare at age 65 and the identification challenges regarding MA enrollment, but they leave open question of whether MA enrollment causally shapes preventive behavior.

This paper provides first quasi-experimental evidence on whether MA enrollment causally increases physical activity among older adults. Using panel data from the Health and Retirement Study from 2012 to 2022, we estimate a Difference-in-Discontinuities design, following Grembi et al. (2016)), that compares the age 65 discontinuity in daily vigorous exercise between MA enrollees and non-MA individuals. The identifying intuition is that while both groups experience the universal shocks age age 65 such as retirement, Social Security claiming, and general aging, only MA enrollees gain access to structured fitness benefit embedded in their plans. Under the assumption that both groups would otherwise experience similar discontinuities in exercise at age 65, the difference in discontinuities isolate the MA-specific behavioral response. Bandwidth sensitivity analysis confirms that the estimates remain stable across a range of local windows around the threshold, and pre-trend evidence supports the identifying assumption.

We find that MA enrollment increases the probability of daily vigorous exercise by approximately 4 percentage points at the Medicare eligibility threshold compared to Non-MA counterpart. The effect is immediate at eligibility but attenuates after two years, suggesting that one-time benefit access generates initial behavioral response but is insufficient for sustained habit formation. This behavioral response is concentrated among groups facing greater structural barrier to exercise. Non-White beneficiaries experience increases in daily exercise of approximately 14 percentage points, while women and individuals without college degress also show meaningfully larger effects than their counterparts.

Our paper makes three contributions to the literature. First, we provide the first quasi-experimental evidence linking MA-specific benefit design to physical activity among older adults which fills gap in the literature on preventive health behavior despite the growing policy relevance of MA supplemental benefits. To our knowledge, this is also first study to examine whether financial incentives for exercise generate behavioral response among older adults as prior literature on financial incentives and physical activity has focused predominantly on younger populations, finding similarly short-lived effects (Charness and Gneezy (2009); Acland and Levy (2015);Royer et al. (2015)) . In addition, we document the behavioral response attenuates after two years of eligibility, contributing to a broader literature in behavioral economics on whether financial incentives generate durable habit formation. Our findings suggest they do not on their own, underscoring the importance of sustained engagement mechanisms beyond initial eligibility. Lastly, the concentration of effects among Non-White, female and less-educated beneficiaries suggests that MA fitness benefits may reduce barriers to exercise that disproportionately affect disadvantaged populations, with the implications for design of supplemental benefits as a tool for narrowing gaps in healthy aging.

2 Data

The main data source is the Health and retirement Study (HRS), a nationally representative longitudinal survey of older adults in the United States conducted biennially by University of Michigan. We use survey waves from 2012 to 2022 providing detailed information on insurance coverage, demographics, labor market status, and self-reported physical activity. As our empirical strategy exploits Medicare eligibility threshold at age 65, a key variable in the analysis is continuous age. We therefore construct a precise age by using the respondent's birth year and month with the survey year and month. This allows us to implement the difference-in-discontinuities design with age as the running variable centered at 65.

The primary outcome measures vigorous physical activity based on the following HRS question: *"How often do you take part in sports or activities that are vigorous, such as running or jogging, swimming, cycling, aerobics or gym workout, tennis, or digging with a spade or shovel?"*. Respondents report in five levels ("Every day," "More than once a week," "Once a week," "One to three times a month," or "Hardly ever or never").

We construct a binary indicator that equals to 1 if the respondent reports daily vigorous workout and 0 otherwise. This threshold is grounded in the 2018 Federal Physical Activity Guidelines for older Americans, which recommend at least 75 to 150 minutes per week of vigorous-intensity aerobic activity combined with additional muscle strengthening exercises. Individuals who engage in vigorous exercise daily are those most likely to meet or exceed this guideline, as even modest daily vigorous activity accumulates well above the weekly minimum over the course of a week. Daily vigorous exercise therefore represents a behaviorally meaningful and policy-relevant threshold that directly corresponds to the type of structured, gym-based activity offered by Medicare Advantage fitness programs.

Beyond its policy relevance, this binary threshold offers two additional econometric advantages. First, it isolates a behaviorally distinct groups of people who are consistent, high-frequency exercisers rather than conflating occasional and habitual exercisers into a noisy continuous measure, providing a clean and interpretable outcome for the discontinuity design. In addition, given the low baseline prevalence of daily vigorous activity among older adults (approximately 3 percent in our sample), even modest absolute increases represent economically meaningful relative changes in behavior. We complement this binary outcome with a quasi-continuous exercise frequency measure — recoding responses as 0 for "hardly ever or never," 1 for "once a week," "one to three times a month," or "more than once a week," and 7 for "every day" — in the robustness checks to confirm that the main results are not driven by the imposed binary cutoff.

In the main analysis, we define insurance groups using observed Medicare coverage histories reported in the HRS. The MA group consists of respondents who are observed in Medicare Advantage but never in Traditional Medicare during the sample period, while the comparison group includes respondents observed in Traditional Medicare or never enrolled in Medicare at the time of the survey. To ensure balanced identification on both sides of the eligibility threshold, we restrict the samples to respondents who are observed in at least one wave before and one wave after age 65. Retaining individuals who are only observed prior to age 65 would contribute observations exclusively to the left side of the cutoff, introducing asymmetry in the local comparison and potentially biasing the discontinuity estimate. This restriction ensures that the DiDC estimate is identified from within-person variation across the eligibility threshold rather than from cross-sectional comparisons between individuals who happen to fall on different sides of age 65 at the time of the survey.

In addition, we drop year 2020 as the COVID-19 pandemic sharply disrupted physical activity opportunities and health behavior such as widespread gym closures due to nationwide lockdown policies that directly affected the fitness benefits at the core of our analysis. We additionally conduct analysis including 2020 separately in robustness checks. Table 1 reports summary statistics for the resulting estimation sample.

Table 1: Baseline Characteristics by Medicare Advantage Status (Ages 60–70)

	Non-MA N(%)	MA N(%)	Total N(%)
Gender			
Male	3,212 (44.13)	1,888 (39.79)	5,100 (42.42)
Female	4,066 (55.87)	2,857 (60.21)	6,923 (57.58)
Race			
White/Caucasian	5,033 (69.49)	2,789 (59.04)	7,822 (65.36)
Black/African American	1,347 (18.60)	1,331 (28.18)	2,678 (22.38)
Other	863 (11.91)	604 (12.79)	1,467 (12.26)
Hispanic			
Not Hispanic	5,984 (82.40)	3,785 (79.87)	9,769 (81.40)
Hispanic	1,278 (17.60)	954 (20.13)	2,232 (18.60)
Education			
No college degree	4,700 (65.05)	3,441 (73.10)	8,141 (68.23)
College degree or higher	2,525 (34.95)	1,266 (26.90)	3,791 (31.77)
Health Conditions			
Health condition index, mean (SD)	1.95 (1.45)	2.06 (1.47)	1.99 (1.46)
Physical Activity (Vigorous)			
Every day	229 (3.16)	145 (3.07)	374 (3.12)
>1 per week	1,903 (26.23)	1,173 (24.80)	3,076 (25.67)
1 per week	840 (11.58)	569 (12.03)	1,409 (11.76)
1–3 per month	815 (11.23)	510 (10.78)	1,325 (11.06)
Never	3,468 (47.80)	2,332 (49.31)	5,800 (48.40)
Total observations	7,278	4,745	12,023

Notes: Counts are shown with column percentages in parentheses for respondents ages 60–70, except for the health condition index, which is reported as mean and standard deviation. The health condition index is based on RAND HRS variable RWCONDE and is defined as the sum of indicators for whether a doctor has ever told the respondent that they had any of the following eight conditions: high blood pressure, diabetes, cancer, lung disease, heart disease, stroke, psychiatric problems, and arthritis. The analytic sample includes 12,023 observations. Sample sizes vary slightly across characteristics due to missing values.

3 Empirical Strategy

A central challenge is that Medicare Advantage enrollment is endogenous. Individuals who select Medicare Advantage over Traditional Medicare or no Medicare may differ systematically in health preferences, socioeconomic status, underlying health conditions, and baseline physical activity. A naive comparison of exercise behavior across insurance groups therefore confounds the causal effect of MA fitness benefit with selection on unobservables.

We address this empirical challenge by using a difference-in-discontinuities(DiDC) design (Grembi et al. (2016)). The key insight is that Medicare eligibility at age 65 generates a sharp, exogenous discontinuity in insurance coverage for all individuals. However, a standard regression discontinuity(RD) design cannot isolate MA-specific fitness benefits because multiple confounding shocks occur contemporaneously at the eligibility cutoff such as retirement, Social Security claiming, and broader changes in aging that may independently affect exercise behavior. Furthermore, plan choices between MA and Traditional Medicare is itself endogenous, as individuals self-select into plans based on their preferences which preclude a clean comparison of health outcomes across two groups. Our strategy addresses this limitation by comparing the magnitude of the age-65 discontinuity in the probability of daily exercise between MA enrollees who gain access to structured fitness benefits, and non-MA beneficiaries, who do not. By differencing out the discontinuities common to both groups, we can isolate the MA-specific behavioral response from these confounders.

Formally, the DiDC estimand is:

$$\begin{aligned}\text{DiDC} = &\Big[E(Workout \mid Post65,\ MA{=}1) - E(Workout \mid Pre65,\ MA{=}1)\Big] \\ &- \Big[E(Workout \mid Post65,\ MA{=}0) - E(Workout \mid Pre65,\ MA{=}0)\Big]. \end{aligned} \tag{1}$$

To clarify what this estimand recovers, we decompose the within-group differences for each insurance group as follows :

$$\Delta^{MA} = \left(Workout^{MA}_{65,\text{post}} - Workout^{MA}_{65,\text{pre}}\right)$$

$$= \underbrace{\text{Universal 65 shock}}_{\text{common to all}} + \text{MA-specific selection} + \underbrace{\text{MA fitness benefit}}_{\text{treatment effect}} + \underbrace{\cancel{\text{Time-invariant selection}}}_{\text{fixed traits}}$$

$$\Delta^{\text{non-MA}} = \left(Workout^{\text{non-MA}}_{65,\text{post}} - Workout^{\text{non-MA}}_{65,\text{pre}}\right)$$

$$= \underbrace{\text{Universal 65 shock}}_{\text{common to all}} + \text{Non-MA-specific selection} + \underbrace{\cancel{\text{Time-invariant selection}}}_{\text{fixed traits}}$$

Within-group differencing cancels time-invariant selection, and taking the difference across groups eliminates the universal age-65 shock common to both groups :

$$\text{DiDC} = \Delta^{MA} - \Delta^{\text{non-MA}} = \text{MA fitness benefit} + (\text{MA-specific selection} - \text{Non-MA-specific selection})$$

The DiDC estimand therefore can removes confounders common to both groups at age 65 as well as time-invariant individual differences in exercise preferences. However, the remaining threat to identification is the possibility that MA and non-MA enrollees already exhibit different exercise trajectories before age 65. to assess this, we perform an event study to check differential pre-trends in exercise behavior prior to age 65. In addition, we conduct the placebo test at a false cutoff of age 66 to verify no spurious discontinuity exists at an age with no institutional change in Medicare benefits as well as donut-hole analysis to check if there is a significant manipulation around the cutoff.

3.1 Difference-in-Discontinuity Specification

We estimate the following reduced-form equation as our main specification:

$$
\begin{aligned}
Y_{it} = \alpha &+ \beta\,(\mathrm{Post}_{it} \times MA_i) + \gamma_1\,\mathrm{Post}_{it} + \gamma_2\,MA_i \\
&+ f(Age_{it}) + \mathrm{Post}_{it}\,f(Age_{it}) + MA_i\,f(Age_{it}) \\
&+ (\mathrm{Post}_{it} \times MA_i)\,f(Age_{it}) + \boldsymbol{\delta}^\top X_{it} + \mu_i + \tau_t + \varepsilon_{it}, \qquad (2)
\end{aligned}
$$

where Y_{it} is a binary indicator equal to 1 if individual i reports daily vigorous exercise at time t, and 0 otherwise. Post_{it} equals 1 if $Age_{it} \geq 65$ and 0 otherwise. MA_i equals 1 for individuals ever observed in Medicare Advantage during the study period. $f(Age_{it})$ is a local-linear function of age, estimated separately on each side of the cutoff and allowed to differ by MA status. X_{it} includes demographic controls for gender, race, and educational attainment. μ_i and τ_t denote individual and year fixed effects, respectively, which absorb time-invariant unobservables and common aggregate shocks. Standard errors are clustered at the individual level.

The coefficient of interest is β, which captures the differential discontinuity at age 65 in the probability of daily vigorous exercise for MA enrollees relative to non-MA individuals. Under the identifying assumption that both groups would have experienced similar discontinuities in exercise at age 65 absent MA-specific fitness benefits, β recovers the reduced-form causal effect of MA fitness benefit access on exercise behavior at the eligibility threshold. The local-linear age controls on each side of the cutoff, interacted with both Post_{it} and MA_i, flexibly absorb smooth age trends in exercise that may differ across groups, ensuring that β is identified from the discontinuous jump at the threshold rather than extrapolation from age trends.

We select the estimation bandwidth using the data-driven procedure of **?**, which selects the bandwidth by minimizing the asymptotic mean squared error of the local polynomial estimator. This approach balances bias and variance in estimation near the threshold without relying on an artifical bandwidth choice. The resulting data-driven bandwidth is approximately 3.5 years. To assess sensitivity, we report estimates across a range of alternative bandwidths and present the full bandwidth sensitivity plot.

To examine the evolution of the treatment effect over the life cycle and formally test for differential pre-trends, we estimate a dynamic DiDC specification. Let b_{it} index one-year age bins relative to age 65, so that $b_{it} = 0$ corresponds to the eligibility year and $b_{it} = -1$ serves as the omitted reference category which is a year before Medicare eligibility (age 64) :

$$\begin{aligned} Y_{it} = \alpha &+ \sum_{k \neq -1} \beta_k \mathbb{1}\{b_{it} = k\}\, MA_i + \sum_{k \neq -1} \phi_k \mathbb{1}\{b_{it} = k\} \\ &+ g(Age_{it}) + \text{Post}_{it}\, g(Age_{it}) + MA_i\, g(Age_{it}) + (\text{Post}_{it} \times MA_i)\, g(Age_{it}) \\ &+ \boldsymbol{\delta}' X_{it} + \mu_i + \tau_t + \varepsilon_{it}. \end{aligned} \tag{3}$$

The coefficients β_k measure the age-bin-specific difference in the probability of daily exercise between MA and non-MA individuals, relative to the reference year $k = -1$. Under the parallel pre-trends assumption, we require $\beta_k \approx 0$ for all $k < 0$, meaning that MA and non-MA enrollees followed parallel trajectories in exercise behavior prior to Medicare eligibility. The post-65 coefficients β_k for $k \geq 0$ trace the timing and persistence of the MA-specific behavioral response, allowing us to assess whether the effect is immediate, delayed, or sustained over time.

4 Results

4.1 Discontinuities in Workout Behaviour

We begin by presenting nonparametric evidence on exercise behavior around the Medicare eligibility threshold. Figure 1 plots linear fit of the monthly binned means of the probability of daily exercise separately for individuals in the Medicare Advantage group and those in the non-MA comparison group. Prior to age 65, MA and non-MA beneficiaries display nearly parallel trends in daily workout likelihood. In addition, exactly at the eligibility threshold, MA enrollees exhibit a discrete increase in daily workout probability, whereas Non-MA enrollees do not. This visual jump anticipates the parametric DiDC estimates and suggests that MA-linked fitness benefits trigger behavioral responses.

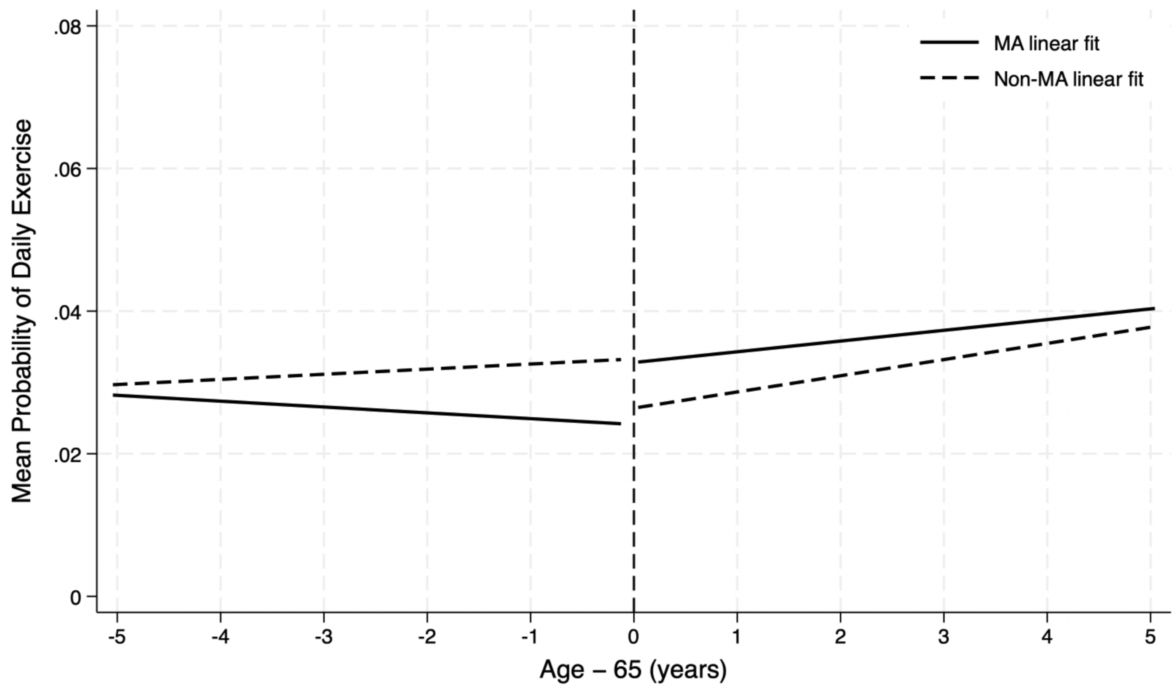

Figure 1
Notes: Binned means of the probability of exercising every day (binary indicator equal to 1 if the respondent reports daily exercise) around age 65 for Medicare Advantage (MA) and non-MA groups. Circles represent MA group means and triangles represent non-MA group means, computed using one-month bins. Solid lines show local linear fits estimated separately on each side of the age-65 cutoff. The vertical dashed line indicates the eligibility threshold at age 65.

Importantly, the figure also demonstrates that while MA enrollees increase their workout levels at age 65, Non-MA drop theirs. This divergent pattern is consistent with a wide range of age-related health, and lifestyle transitions such as retirement, increased caregiving responsibilities, and broader aging related declines in physical activity. The presence of these countervailing forces at age 65 emphasize why neither a simple RD nor a standard DiD design can isolate the causal effect of MA on physical activity.

A standard RD at age 65 would conflate MA-specific behavioral responses with universal age-65 shocks while a traditional DiD would not effectively control non-linear age profiles or discontinuities common to both groups. The DiDC framework addresses both limitations by explicitly differencing out the age-65 discontinuity experience by both groups, allowing us to attribute the residual discontinuity among MA enrollees specifically to MA fitness incentives rather than to broader aging effect.

4.2 Difference-in-Discontinuities results

Figure 2 presents the first stage discontinuity in Medicare Advantange enrollment at the Medicare eligibility cutoff. Consistent with the institutional setting, Medicare Advantage enrollment increases sharply at age 65, indicating that Medicare eligibility generates substantial variation. This pattern supports the empirical relevance of the RD design and confirms that age 65 provides a meaningful source of identifying exogenous variation.

Table 2 reports reduced-form DiDC estimates of the differential age-65 discontinuity in the probability of daily vigorous exercise for the MA group relative to the non-MA group across several bandwidth choices. Our preferred specification uses a data-driven bandwidth selection procedure following Calonico et al. (2014). Specifically, we adopt the mean squared error bandwidth, denoted by b, as the primary local window around the eligibility cutoff. In our setting, the selected reference bandwidth is approximately 3.5 years. This approach has the advantage of allowing the local estimation window to be determined by the statistical properties of the data rather than by an arbitrary bandwidth choice, thereby balancing bias and variance in estimation near the threshold.

At this data-driven bandwidth, the estimated differential discontinuity implies that MA enrollment increases the probability of daily vigorous exercise by about 4 percentage points relative to non-MA beneficiaries. Although this binary outcome does not directly measure total weekly minutes of exercise, the magnitude is economically meaningful given the low baseline prevalence of daily vigorous activity among older adults. More broadly, the estimate suggests that MA-linked supplemental fitness benefits may induce a measurable increase in preventive health behavior at the point of Medicare eligibility.

To assess sensitivity to the choice of local window, Panel B of Table 2 reports estimates using narrower and wider bandwidths around the cutoff, and Figure **??** plots the corresponding point estimates and 95% confidence intervals across bandwidths. This is particularly relevant in the Health and Retirement Study, which is conducted biennially, since bandwidths of roughly two to four years naturally correspond to comparisons using one or two survey waves on either side of age 65. The figure shows that the estimate obtained using the narrowest window is relatively noisy, reflecting limited precision in a very local sample. In contrast, for bandwidths of 2.5 years and above, the estimates remain positive and fairly stable, clustering around the main specification. Taken together, the table and figure indicate that the main result is not driven by a single bandwidth choice, but instead reflects a stable positive pattern in the data around the age-65 cutoff.

Figure 3 presents the event study of the DiDC design, which we use primarily for assessing the parallel pre-65 trend which is the critical assumption of our identification strategy. The difference in discontinuities between MA and non-MA beneficiaries are flat and statistically insignificant which supports the key assumption that, in the absence of MA-specific fitness benefits, both groups would have shown similar discontinuities in the likelihood of daily exercise. At the Medicare eligibility threshold, the event study plot shows a sharp jump in physical activity among MA enrollees relative to non-MA enrollees. The magnitude is considerable and immediate precisely at age 65 which aligns with the activation of MA-linked fitness benefit. However, this effect attenuates and becomes insignificant overtime. This highlights a short-run behavioral boost as individuals gain access to free fitness benefits, consistent with the idea that new financial incentives initially increase the workout frequency.

This attenuation in later years aligns with existing evidence in the physical activity literature among younger population (Charness and Gneezy (2009); Acland and Levy (2015);Royer et al. (2015)) , which has shown that financial incentives, fitness subsidies generate short-run and immediate response but it is hard to sustain this positive impact in the long-run. Commitment, habit formation and motivation, rather than financial incentives alone, are the key barriers to long-term adherence in physical activity.

To further address the concern that MA and Non-MA enrollees may already differ in workout level prior to Medicare eligibility, we complement our main DiDC specification with standard event study analysis. Figure 4 presents regular event study comparing raw workout levels (not discontinuities) between MA and non-MA groups. This plot reveals that both groups show similar workout levels and parallel trends prior to Medicare eligibility, with no evidence of systematic divergence in pre-treatment periods. Further, at age 65, we observe a considerable gap with MA beneficiaries increasing their workout frequency while non-MA individuals show no comparable change. However, we do not interpret these post-65 coefficients as causal estimates since enrollment in Medicare Advantage versus enrollment in Traditional Medicare are both endogenous choices.

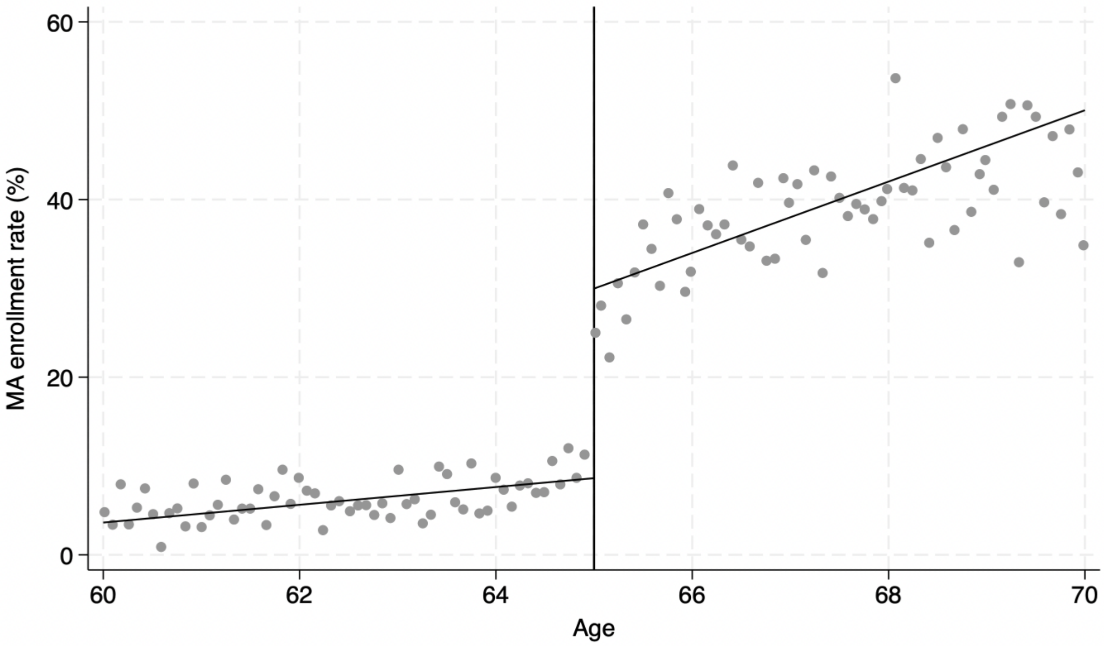

Figure 2
First stage Discontinuity in Medicare Advantage Enrollment at Age 65

Table 2: DiDC Estimates of the Differential Jump in Daily Exercise at Age 65

	Panel A: Main	Panel B: Alternative Bandwidths			
	(1) $h = 3.53$	(2) $h = 1$	(3) $h = 2$	(4) $h = 4$	(5) $h = 5$
$\Delta^{MA} - \Delta^{\text{Non-MA}}$ (pp)[a]	0.040***	0.441*	0.033	0.031**	0.020
	(0.015)	(0.247)	(0.030)	(0.014)	(0.012)
Pre-65 mean, MA group[b]	0.026	0.015	0.022	0.025	0.025
Post-65 mean, MA group[b]	0.037	0.029	0.039	0.038	0.038
Pre-65 mean, Non-MA group[b]	0.031	0.040	0.026	0.031	0.030
Post-65 mean, Non-MA group[b]	0.033	0.024	0.025	0.033	0.033
Observations	8,718	384	3,816	10,009	11,891
R^2	0.498	0.505	0.574	0.476	0.432
Individual fixed effects	✓	✓	✓	✓	✓
Year fixed effects	✓	✓	✓	✓	✓

Notes: Table reports reduced-form DiDC estimates of the differential discontinuity at age 65 in a binary indicator for daily exercise, using different bandwidth choices h (in years). The dependent variable equals 1 if the respondent reports exercising "every day", and 0 otherwise. The coefficient $\Delta^{MA} - \Delta^{\text{Non-MA}}$ captures the difference between the jump at age 65 for individuals in the MA group and the corresponding jump for the non-MA comparison group.[a] Coefficients are reported in percentage points.[b] Means are computed within the estimation sample for each bandwidth, separately by MA status and by whether age is below (Pre-65) or above (Post-65) the cutoff; standard errors clustered at the individual level are reported in parentheses. All specifications include individual and year fixed effects. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

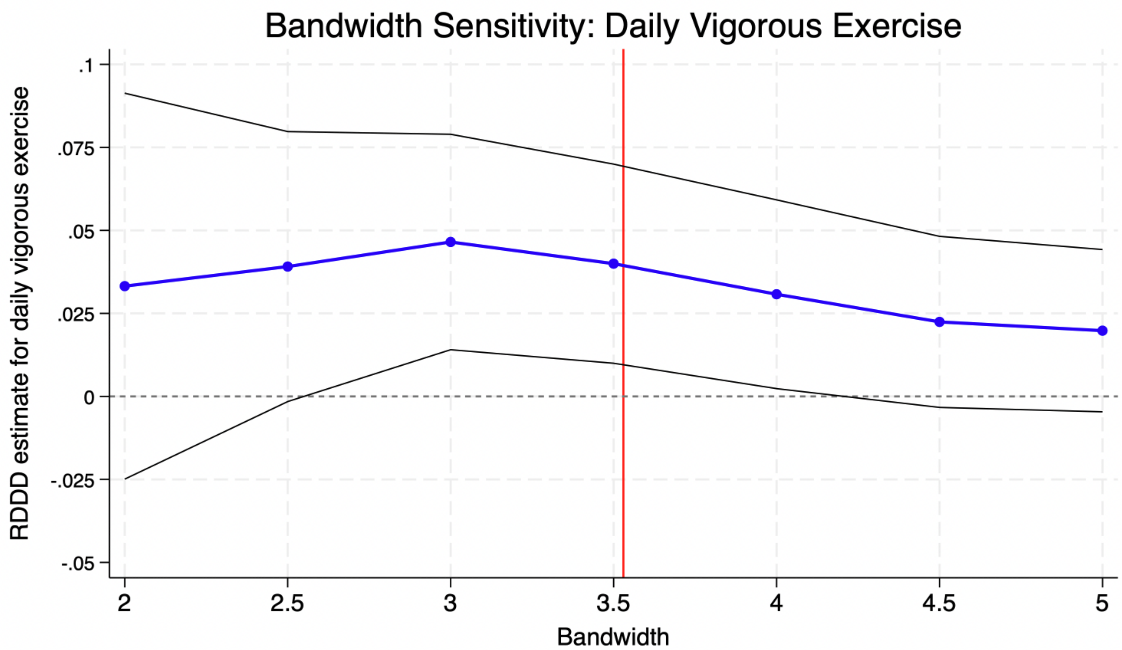

Figure 3
Bandwidth Sensitivity of the Difference-in-Discontinuity Estimate. The figure plots the estimated DiDC coefficient for daily vigorous exercise across alternative bandwidths around the age-65 cutoff. The solid line denotes the point estimate and the surrounding lines denote 95% confidence intervals. The vertical line marks the preferred data-driven bandwidth of 3.53 years.

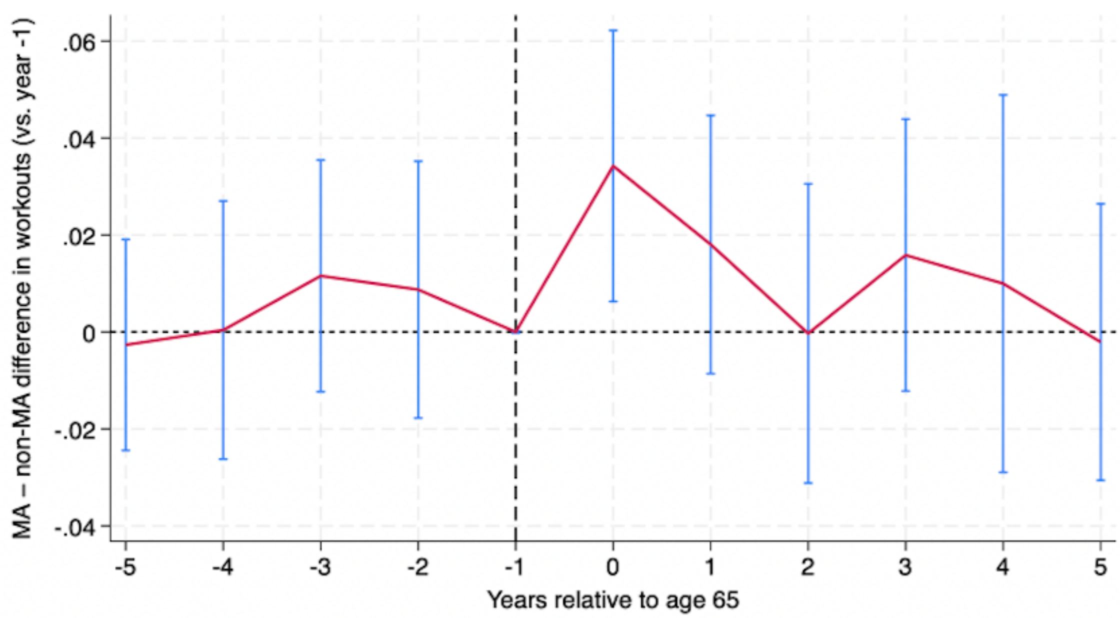

Figure 4
Notes: Event-study estimates of yearly differences in workout level between Medicare Advantage (MA) and Non-MA beneficiaries, relative to a reference year. The dotted horizontal line indicates zero difference and the dashed vertical line marks a reference year which is a year before Medicare eligibility. Points represent bin means and bars show 95% confidence intervals.

5 Heterogeneous Analysis

We investigate whether the DiDC differential effect of medicare eligibility on daily exercise differs across subgroups. Heterogeneity analysis is crucial for our understanding of how behavioral response to Medicare Advantage's fitness benefits. We examine heterogeneity along three dimensions : Gender, Race/Ethnicity (White vs all other groups) , Educational attainment (College degree vs No College degree). For each subgroup analysis, we estimate the same reduced-form DiDC specification used in the main analysis and report static DiDC estimate using data-driven optimal bandwidths and alternative bandwidths as well as dynamic event study.

The heterogeneity results presented in Figure 5 underscore the meaningful variations in behavioral responses to Medicare Advantage (MA) fitness benefits at age 65. Female beneficiaries experience an increase of approximately 5–6 percentage points in the probability of exercising daily, while the corresponding estimate for males is smaller and statistically insignificant. Racial difference is particularly conspicuous where Non-White beneficiaries experience a large increase of approximately 14 percentage points. We do not detect any significant impact among White beneficiaries. Educational differences mirror this pattern as well. College-educated respondents show small and statistically insignificant effects, while non-college educated seniors show statistically significant increases of roughly 5 percentage points. Overall, we find meaningful impact among female, Non-White and non-college educated beneficiaries.

This evidence emphasizes MA supplemental fitness benefits may generate larger gains among populations that historically face greater structural barriers to physical activity which speaks to equitable healthcare access. Non-White and lower-educated seniors often face financaial and informational frictions, limited access to recreational facilities, and higher baseline health risks. The substantial effects among disadvantaged groups is consistent with the interpretation that MA fitness incentives may help reduce disparities in preventive health behaviors at later life. Even though the analysis does not necessarily measure the inequality reduction, the distribution of estimated impacts aligns with an equity enhancing mechanisms

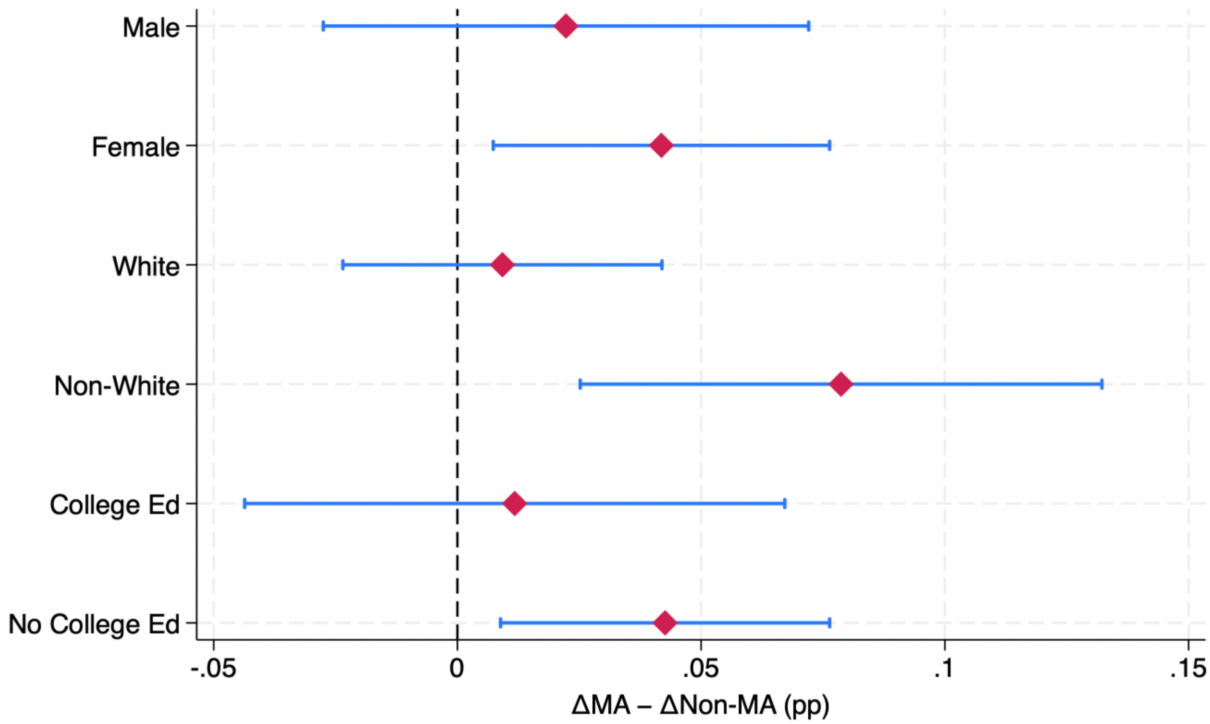

Figure 5
Notes: Reduced-form regression discontinuity in difference-in-discontinuities (DiDC) estimates of the differential jump at age 65 in daily exercise across demographic subgroups. Points denote subgroup-specific estimates of $\Delta^{MA} - \Delta^{\text{Non-MA}}$ expressed in percentage points. Horizontal bars represent 95% confidence intervals based on standard errors clustered at the individual level. Estimates are computed using the MSE-optimal bandwidth and include individual and year fixed effects.

6 Robustness Check

To assess the validity of our results, we perform a comprehensive set of robustness checks that address potential concerns about our findings. These series of robustness checks support the estimated differential discontinuity at age 65 represents a genuine behavioral response associated with the enrollment in Medicare Advantage with fitness benefits, rather than the artifacts of model specification, age related dynamics,and local irregularities in the running variables, or spurious discontinuities.

6.1 Alternative Outcomes: Retirement, Private Insurance Enrollment, Alternative Exercise Measure, and Moderate Physical Activity

We estimate the same DiDC specification using several alternative outcomes that may change around Medicare eligibility and potentially affect workout frequencies: retirement, private insurance coverage, an alternative measure of vigorous exercise, and moderate physical activity. The purpose of these tests is to rule out the possibility that the estimated physical activity jump reflects broader life-cycle transitions around age 65. Consistent with expectations, retirement does not show a meaningful differential discontinuity at age 65, as shown in Table 3. In addition, we find a statistically significant decline in the number of private insurance plans at age 65 for the MA group relative to the non-MA group, as shown in Table 3. This sharp reduction indicates that seniors are not simultaneously holding private coverage that could independently encourage exercise. The only insurance margin that changes discontinuously at age 65 for the MA group is the shift into MA plans, which uniquely offer structured free fitness benefits.

We also examine whether the main result is robust to using an alternative measure of vigorous physical activity that preserves more information from the original survey categories. Specifically, we construct an exercise-frequency score equal to 7 for respondents who report vigorous activity "every day," 1 for respondents who report vigorous activity more than once per week, once per week, or one to three times per month, and 0 for respondents who report never engaging in vigorous activity. As shown in Table 3, the estimated differential discontinuity remains positive under this alternative coding. The estimate is marginally significant under the main bandwidth and statistically significant under the $h = 3$ specification, although the magnitude and precision vary across bandwidths. Because this outcome is coded as an exercise-frequency score rather than a binary indicator, the coefficient should be interpreted in score units rather than percentage points. Overall, this pattern provides additional support that MA enrollment is associated with increased vigorous exercise behavior around age 65.

To assess whether the estimated MA effect reflects an increase in overall physical activity rather than a mechanism linked to structured fitness benefits, we examine a separate outcome measuring daily engagement in moderate physical activities such as gardening, walking at a moderate pace, dancing, and car washing. Table 3 shows that the differential discontinuity at age 65 in this outcome is small and statistically insignificant across all bandwidth choices. The absence of a significant effect on daily moderate activity suggests that our primary findings on vigorous physical activity are unlikely to be driven by broad shifts in activity or general health motivation triggered by Medicare eligibility. Instead, the results are consistent with a more targeted mechanism in which MA-related fitness benefits primarily affect structured or vigorous exercise behaviors, such as gym-based workouts, rather than everyday moderate physical activity.

Table 3: DiDC Estimates of Differential Jumps at Age 65 Across Outcomes

	Alternative Outcomes			
	Retirement	Private plan	Moderate PA	Alternative workout
	(1)	(2)	(3)	(4)
	$b = 3.530$	$b = 3.530$	$b = 4.541$	$b = 3.820$
$\Delta^{MA} - \Delta^{\text{Non-MA}a}$	0.043	-0.379***	0.019	0.184*
	(0.029)	(0.043)	(0.021)	(0.098)
Mean dependent variable	0.617	0.634	0.095	0.696
Pre-65 mean, MA group[b]	0.566	0.703	0.070	0.658
Post-65 mean, MA group[b]	0.818	0.181	0.099	0.704
Pre-65 mean, Non-MA group[b]	0.499	0.772	0.090	0.710
Post-65 mean, Non-MA group[b]	0.678	0.687	0.122	0.704
Observations	8,556	8,715	11,060	9,418
R^2	0.754	0.688	0.455	0.556
Individual fixed effects	✓	✓	✓	✓
Year fixed effects	✓	✓	✓	✓

Notes: This table reports difference-in-discontinuities (DiDC) estimates of the differential jump at age 65 for several alternative outcomes. Each column uses the preferred bandwidth selected for the corresponding outcome. The dependent variables are retirement status, private plan coverage, daily moderate physical activity, and the alternative workout-frequency measure. The coefficient $\Delta^{MA} - \Delta^{\text{Non-MA}}$ reports the differential discontinuity for the MA group relative to the non-MA group at age 65. Pre- and post-65 means are computed within the estimation sample by MA status and age group. Standard errors clustered at the individual level are reported in parentheses. All specifications include individual and year fixed effects. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

6.2 Robustness checks

We conduct three additional robustness checks to assess whether the main DiDC estimate is driven by local irregularities around the age-65 cutoff, by spurious age-related discontinuities, or by the exclusion of COVID-era observations. First, we estimate a donut-hole specification that excludes observations within three months on either side of age 65. This test addresses concerns that very local administrative errors, reporting issues, or mechanical changes immediately around the Medicare eligibility threshold may drive the results. Second, we estimate a placebo DiDC specification using age 66 as a false cutoff. Because age 66 does not correspond to the main Medicare eligibility threshold, we should not observe a meaningful differential discontinuity at this age if the main design is valid. Third, we re-estimate the main specification including the COVID-era observations that are excluded from the preferred specification.

Table 4 reports the results using the reference bandwidth for each robustness exercise. The donut-hole estimate remains positive and similar in magnitude to the main estimate, suggesting that the result is not driven by observations immediately around the age-65 cutoff. The placebo estimate at age 66 is small, negative, and statistically insignificant, which supports the interpretation that the main discontinuity is specific to the true Medicare eligibility threshold. Finally, when COVID-era observations are included, the estimate remains positive and statistically significant, indicating that the main finding is not mechanically driven by excluding the COVID period.

Table 4: Robustness Checks

	Donut hole	Placebo cutoff	Including COVID
	(1) $h = 3.53$	(2) $h = 4.75$	(3) $h = 3.58$
$\Delta^{MA} - \Delta^{\text{Non-MA}}$	0.038**	-0.009	0.032**
	(0.016)	(0.014)	(0.013)
Pre-cutoff mean, MA group	0.026	0.027	0.024
Post-cutoff mean, MA group	0.037	0.036	0.035
Pre-cutoff mean, Non-MA group	0.031	0.028	0.032
Post-cutoff mean, Non-MA group	0.034	0.038	0.032
Observations	8,611	10,385	9,941
R^2	0.186	0.457	0.466
Individual fixed effects	✓	✓	✓
Year fixed effects	✓	✓	✓

Notes: The table reports reduced-form DiDC robustness estimates using the reference bandwidth from each specification. Column (1) reports a donut-hole specification that excludes observations within one months on either side of age 65. Column (2) reports a placebo specification using age 66 as the false cutoff. Column (3) reports the main specification after including COVID-era observations. The dependent variable equals one if the respondent reports exercising every day, and zero otherwise. Means are computed within the estimation sample for each specification. Standard errors clustered at the individual level are reported in parentheses. All specifications include individual and year fixed effects. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

6.3 Fitness benefit coverage over time

The proportion of Medicare Advantage which covers fitness benefits has expanded substantially during the late 2010s and early 2020s. KFF (2022)reports that in 2022, 97% of individual Medicare Advantage plans offers a fitness benefit, underscoring the near-universal availability of fitness-related supplemental coverage in recent years. We assess whether the impact of MA covered fitness benefit is more pronounced in later years when fitness benefits are more widely available.

For this analysis, we split the analysis into an early period (2012-2014) and a later period (2016–2022). Table **??** shows that estimates in 2012-2014 are small and statistically insignificant across all bandwidth choices, with even negative impact at the optimal bandwidth. In contrast, Table **??** shows more positive, bigger, and statistically significant effects in 2016–2022 for several bandwidths. While the optimal-bandwidth estimate in the later period is not statistically significant, precision is sensitive to bandwidth because the effective estimation sample shrinks substantially at narrow bandwidths. Overall, this earlier year versus later year results are consistent with stronger behavioral responses in the later period when fitness benefits were more broadly available in Medicare Advantage plans.

Table 5: DiDC Estimates of the Differential Jump in Daily Exercise at Age 65 by Period

	Dependent Variable: Daily Exercise	
	2012–2014 (1) $h = 1.465$	2016–2022 (2) $h = 4.100$
$\Delta^{MA} - \Delta^{\text{Non-MA}}$[a]	0.046 (0.122)	0.038* (0.023)
Pre-65 mean, MA group	0.009	0.029
Post-65 mean, MA group	0.018	0.042
Pre-65 mean, Non-MA group	0.020	0.034
Post-65 mean, Non-MA group	0.035	0.037
Observations	630	6,045
R^2	0.567	0.538
Individual fixed effects	✓	✓
Year fixed effects	✓	✓

Notes: Columns report reduced-form DiDC estimates using the preferred data-driven bandwidth for each period. Standard errors clustered at the individual level are in parentheses. All specifications include individual and year fixed effects. [a] The coefficient $\Delta^{MA} - \Delta^{\text{Non-MA}}$ captures the difference between the jump at age 65 for individuals in the MA group and the corresponding jump for the non-MA comparison group. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

7 Conclusion

We provide the first quasi-experimental evidence on how Medicare Advantage (MA) enrollment shapes seniors' vigorous exercise with the binary indicator for exercising daily. Exercising daily gets higher chance for seniors to meet the federal physical activity guideline which is more than 150-300 minutes of moderate-intensity , or 75-150 minutes vigorous-intensity exercise per week. Exploiting the Difference-in-Discontinuities(DiDC), we estimate a reduced-form of differential discontinuity in the probability of exercising everyday at age 65 for MA enrollees compared to Non-MA enrollees. This approach addresses selection concerns by differencing out both observable and unobservable characteristics within and across groups. Using nationally representative panel data from Health and Retirement Study, we find a meaningful and sizable increase in the probability of daily exercise at age 65 for the MA group compared to Non-MA group. These results emphasize the behavioral impact of supplemental fitness benefit which is uniquely provided by MA plans.

Our findings have several implications and contributions. First, they demonstrate health insurance benefit structure trigger behavioral response, preventive health behavior in particular. By reducing financial and logistical barriers to structured exercise programs, MA plans effectively lead to more active lifestyles when the free fitness benefits activate at the Medicare eligibility. Even modest improvements in vigorous physical activity at older ages may generate substantial effect considering the fact that the baseline physical activity is significantly low among older populations.

In addition, the heterogeneity results underscore who benefits the most from these incentives. The largest increase in daily exercise occur among Non-White senior groups that historically face structural barriers to physical activity and gym access. Our finding signals that fitness benefits embedded in MA plans may be particularly effective in narrowing socioeconomic and racial disparities in healthy aging especially through improving the access to better physical activity.

Lastly, the attenuation of effects after age 65 demonstrates the need for sustained engagement. While MA eligibility induces an immediate behavioral response, maintaining improvements in physical activity requires recurring incentives, targeted outreach, government support, or even more personalized wellness program. Our results call attention to the potential value of continued reinforcement rather than one-time eligibility shock.

As MA penetration continues to grow exponentially, the scope and the structure of supplemental fitness benefit will play an increasingly crucial role in shaping health trajectory of the aging population. Our findings emphasize the well-designed, structured insurance programs can function as a policy lever that actively promote healthy aging and provide actionable guidance for policymakers and insurers seeking to design programs that foster healthy life-style.

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